Chapter 7F

Hematology/Oncology Review - Integrated Cases, Diagnostic Algorithms, and Graph-RAG Map

A content-heavy capstone chapter synthesizing CBC interpretation, anemia, hemolysis, coagulation, transfusion, hematologic malignancy, solid oncology, oncologic emergencies, treatment toxicity, and integrated clinical reasoning.

How to use this chapter

Read this as a diagnostic control panel rather than a memorization list. Each syndrome is framed by the same sequence: first stabilize, then classify by objective data, then connect the abnormality to mechanism, then choose the most likely disease process, then anticipate complications. The goal is to make the next step in evaluation obvious from a small set of lab and clinical anchors.

1. The Core Hematology/Oncology Problem List

Hematology and oncology cases often begin with nonspecific language: fatigue, bruising, weight loss, recurrent infection, dyspnea, lymphadenopathy, bone pain, abnormal bleeding, abnormal clotting, or an incidental CBC abnormality. The first task is not to name a disease. It is to decide which physiologic compartment has failed: red cell mass and oxygen delivery, white cell number or function, platelet number or function, plasma coagulation factors, marrow architecture, lymphoid tissue, plasma cell immunoglobulin production, or solid tumor behavior. Each compartment leaves a different fingerprint in the CBC, smear, reticulocyte response, coagulation tests, chemistry panel, and physical exam.

Clinical starting point	First interpretation	Immediate danger	High-yield first data
Fatigue, pallor, dyspnea	Anemia or systemic cancer burden	Symptomatic hypoxia, coronary ischemia, occult bleeding	Hb/Hct, MCV, reticulocyte index, smear, ferritin/iron, B12/folate, hemolysis labs
Petechiae, mucosal bleeding	Platelet number/function problem or vWD	Intracranial/GI bleeding if severe thrombocytopenia	Platelet count, smear, PT/PTT, fibrinogen, D-dimer, medication review
Hemarthrosis, deep muscle hematoma	Coagulation factor problem	Compartment bleeding or CNS bleeding	PT, PTT, mixing study, factor levels, vWF testing
Thrombosis with low platelets	Consumptive or immune platelet-clotting syndrome	HIT, TTP, DIC, antiphospholipid syndrome	Timing, heparin exposure, smear for schistocytes, hemolysis labs, fibrinogen/D-dimer
Lymphadenopathy/B symptoms	Lymphoma, leukemia, infection, autoimmune disease	Airway compromise, SVC syndrome, tumor lysis risk	Node exam, CBC/diff, LDH, imaging, excisional biopsy if persistent
Bone pain, renal failure, anemia, hypercalcemia	Plasma cell disorder until proven otherwise	Renal failure, pathologic fracture, hyperviscosity	SPEP/UPEP, immunofixation, free light chains, calcium, creatinine, skeletal imaging

2. Universal Stabilization Before Classification

Before building a differential, identify whether the patient is unstable. A hemodynamically unstable patient with suspected bleeding should be resuscitated before the complete anemia workup. A patient with neutropenic fever should receive empiric antipseudomonal antibiotics early, not after every test returns. A patient with TTP features should be treated urgently with plasma exchange while confirmatory ADAMTS13 testing is pending. A patient with leukostasis, tumor lysis syndrome, malignant spinal cord compression, or symptomatic hypercalcemia needs emergency-directed therapy before a fully staged cancer plan is available.

Emergency screen in any heme/onc case

Check ABCs, mental status, perfusion, fever, active bleeding, chest pain, dyspnea, neurologic deficits, and severe pain. | Obtain CBC with differential, reticulocyte count if anemic, smear, CMP, PT/INR, PTT, fibrinogen, D-dimer if coagulopathy suspected, type and screen if bleeding, LDH/uric acid/phosphate/calcium if malignancy or tumor lysis is possible. | Treat immediately when a time-sensitive syndrome is likely: sepsis/neutropenic fever, DIC with bleeding, TTP, HIT with thrombosis, acute leukemia with leukostasis, tumor lysis, spinal cord compression, SVC syndrome, symptomatic hypercalcemia, or severe anemia with ischemia. | After stabilization, classify with the smallest reliable algorithm: reticulocyte response for anemia, platelet count plus PT/PTT for bleeding, smear plus hemolysis labs for MAHA, node biopsy for lymphoma, SPEP/UPEP/free light chains for plasma cell disease.

3. CBC Interpretation as a Diagnostic Map

The CBC is a structured map of marrow output and peripheral destruction. Hemoglobin describes red cell oxygen-carrying capacity; MCV describes the average size of red cells; RDW describes size variability; the white blood cell differential separates neutrophilic, lymphocytic, eosinophilic, monocytic, and blast-predominant processes; and the platelet count estimates primary hemostatic capacity. No single number is enough. A platelet count of 40,000/uL after heparin is a thrombosis warning; the same count after chemotherapy may be a bleeding warning. A WBC of 60,000/uL with mature neutrophils and toxic granulation suggests infection or inflammation, whereas a WBC of 60,000/uL with blasts suggests leukemia.

CBC pattern	Mechanistic reading	Most useful next distinction	Common examples
Low Hb with high reticulocytes	Marrow is responding; loss/destruction likely	Blood loss vs hemolysis	GI bleeding, menstruation, autoimmune hemolysis, sickle crisis, G6PD deficiency
Low Hb with low/normal reticulocytes	Inadequate production	MCV and nutritional/inflammatory/marrow clues	Iron deficiency, B12/folate deficiency, CKD, anemia of inflammation, aplastic anemia, marrow infiltration
Pancytopenia	Global marrow output failure or sequestration	Hypocellular vs infiltrated marrow; splenomegaly	Aplastic anemia, myelodysplasia, leukemia, myelofibrosis, hypersplenism
Neutropenia with fever	Impaired innate defense	Chemotherapy vs marrow failure vs drug/infection	Febrile neutropenia, aplastic anemia, leukemia, medication toxicity
Lymphocytosis	Reactive or clonal lymphoid expansion	Age, smear, symptoms, flow cytometry	Viral infection, CLL, lymphoma spillover, ALL
Thrombocytopenia isolated	Platelet destruction/sequestration likely if other lines normal	Immune/drug/HIT/TTP/DIC vs pseudothrombocytopenia	ITP, HIT, TTP, viral, medications
Thrombocytopenia plus anemia with schistocytes	Microangiopathic hemolysis	PT/PTT/fibrinogen and renal/CNS pattern	TTP, HUS, DIC, malignant hypertension, mechanical valve

4. Anemia Algorithm: Production Failure vs Loss/Destruction

The most efficient anemia algorithm begins with hemodynamic stability and reticulocyte response. A low reticulocyte index means the marrow is not keeping up; an elevated reticulocyte index means the marrow recognizes anemia and is attempting compensation. MCV then narrows production failure, but MCV is not destiny: early iron deficiency may be normocytic, anemia of inflammation can be normocytic or microcytic, reticulocytosis can raise MCV, and mixed iron plus B12 deficiency can normalize the MCV while the RDW rises.

Anemia decision pathway

Confirm true anemia and evaluate severity: Hb/Hct, symptoms, vitals, evidence of active bleeding or ischemia. | Reticulocyte index elevated (>~2): think blood loss or hemolysis. Order LDH, indirect bilirubin, haptoglobin, DAT/Coombs, smear, urinalysis, and bleeding evaluation. | Reticulocyte index low/inappropriately normal: classify by MCV. Microcytic suggests iron restriction or globin/heme synthesis disease. Macrocytic suggests megaloblastic DNA synthesis failure, liver disease, alcohol, hypothyroidism, medication, or reticulocytosis. Normocytic suggests CKD, inflammation, marrow failure/infiltration, early nutritional deficiency, or endocrine disease. | Use smear morphology to correct the algorithm: target cells suggest thalassemia/liver disease; hypersegmented neutrophils suggest megaloblastic anemia; schistocytes suggest MAHA; spherocytes suggest immune hemolysis or hereditary spherocytosis; sickled cells suggest sickle disease; rouleaux suggests paraproteinemia. | Treat the cause, not the MCV. Iron deficiency in an older adult is GI blood loss until proven otherwise; B12 deficiency requires neurologic protection; anemia of CKD requires kidney and iron context; hemolysis requires identifying immune, mechanical, intrinsic RBC, infectious, or microangiopathic drivers.

Anemia type	Mechanism	Key labs/smear	High-yield clinical link	Common pitfall
Iron deficiency	Insufficient iron for heme synthesis	Low ferritin, low iron, high TIBC, high RDW, microcytosis over time	Pica, restless legs, menstrual or GI blood loss	Giving iron without searching for bleeding in men/postmenopausal women
Anemia of inflammation	Hepcidin-mediated iron sequestration and suppressed erythropoiesis	Low iron, low/normal TIBC, normal/high ferritin	Chronic infection, autoimmune disease, malignancy, CKD	Mistaking high ferritin for iron overload when inflammation is present
Thalassemia	Reduced alpha or beta globin production	Microcytosis often disproportionate to anemia; target cells; normal/high RBC count	Family/ethnic background; poor response to iron	Treating indefinitely as iron deficiency

Anemia type	Mechanism	Key labs/smear	High-yield clinical link	Common pitfall
Sideroblastic anemia	Defective heme synthesis with mitochondrial iron trapping	Ring sideroblasts; iron/ferritin often high	Alcohol, lead, isoniazid, myelodysplasia, copper deficiency	Assuming every microcytic anemia needs iron
B12 deficiency	Impaired DNA synthesis and neurologic myelin injury	Macrocytosis, hypersegmented neutrophils, high MMA and homocysteine	Pernicious anemia, ileal disease, vegan diet, metformin/PPI association	Giving folate alone and missing irreversible neurologic injury
Folate deficiency	Impaired DNA synthesis without myelin injury	Macrocytosis, high homocysteine, normal MMA	Alcohol use, malnutrition, pregnancy, methotrexate, hemolysis	Not ruling out B12 deficiency before treatment
Aplastic anemia	Hypocellular marrow failure	Pancytopenia, low reticulocytes, hypocellular marrow	Drugs, toxins, viral hepatitis, idiopathic, inherited marrow failure	Missing infection/bleeding risk while pursuing diagnosis

5. Hemolysis and Hemoglobinopathy Integration

Hemolysis is not just anemia; it is premature red cell destruction. The marrow usually responds with reticulocytosis unless the marrow is suppressed or substrate-limited. The classic biochemical pattern is elevated LDH, elevated indirect bilirubin, reduced haptoglobin, and reticulocytosis. The smear determines whether destruction is immune, mechanical, membrane-related, enzyme-related, hemoglobin-related, or microangiopathic. Intravascular hemolysis tends to produce hemoglobinemia, hemoglobinuria, and very low haptoglobin; extravascular hemolysis occurs mainly through splenic or hepatic macrophage clearance and often presents with jaundice and splenomegaly.

Smear clue	Likely process	Mechanism	Confirmatory tests / next move
Spherocytes	Warm autoimmune hemolytic anemia or hereditary spherocytosis	Loss of membrane surface area; splenic clearance	DAT/Coombs for AIHA; osmotic fragility or EMA binding for hereditary spherocytosis
Schistocytes	Microangiopathic or mechanical hemolysis	RBC shearing through fibrin, damaged endothelium, or prosthetic valve	Platelets, PT/PTT/fibrinogen, creatinine, neurologic signs; treat TTP/DIC/HUS urgently
Bite cells / Heinz bodies	G6PD deficiency	Oxidant injury denatures hemoglobin	G6PD assay after acute episode; avoid oxidant drugs/foods
Sickled cells	Sickle cell disease	HbS polymerization during deoxygenation	Hemoglobin electrophoresis; manage pain, infection, acute chest, stroke risk
Target cells	Thalassemia, liver disease, hemoglobin C	Excess membrane relative to hemoglobin or abnormal hemoglobin	Iron studies and hemoglobin electrophoresis
Rouleaux	Plasma cell dyscrasia or inflammation	High plasma protein reduces RBC repulsion	SPEP/UPEP, immunofixation, free light chains

Sickle cell disease deserves a special integrated view because it creates vaso-occlusion, chronic hemolysis, functional asplenia, infection risk, stroke risk, pulmonary complications, renal concentrating defects, avascular necrosis, gallstones, priapism, leg ulcers, and pregnancy risk. The immediate question in a sickle patient is whether the event is an uncomplicated pain episode or a dangerous phenotype such as acute chest syndrome, stroke, splenic sequestration, aplastic crisis, sepsis, or severe anemia requiring transfusion or exchange transfusion.

Sickle complication	Clinical anchor	Mechanism	Management concept
Vaso-occlusive pain	Severe limb/back/chest pain without alternate source	Microvascular occlusion and inflammation	Analgesia, hydration without overload, oxygen only if hypoxemic, evaluate triggers
Acute chest syndrome	New pulmonary infiltrate plus fever, chest pain, cough, hypoxemia	Infection, fat embolism, infarction, hypoventilation	Antibiotics, oxygen, incentive spirometry, transfusion/exchange in severe disease
Stroke	Focal neurologic deficit	Large-vessel or microvascular sickling	Emergency transfusion/exchange, neuroimaging, secondary prevention
Aplastic crisis	Abrupt anemia with low reticulocytes	Parvovirus B19 marrow suppression	Supportive transfusion, infection precautions
Splenic sequestration	Acute splenomegaly, hypovolemia, falling Hb	Pooling of blood in spleen	Volume and transfusion support; often pediatric
Functional asplenia	Encapsulated organism infection risk	Repeated splenic infarction	Vaccines, fever evaluation, prophylaxis in children

6. Bleeding and Clotting: Platelets vs Coagulation Factors

Bleeding pattern often localizes the problem. Platelet disorders cause mucocutaneous bleeding: petechiae, purpura, epistaxis, gum bleeding, menorrhagia, and immediate bleeding after procedures. Coagulation factor disorders cause deep bleeding: hemarthroses, large hematomas, retroperitoneal bleeding, delayed surgical bleeding, and intracranial hemorrhage. DIC, liver disease, anticoagulants, massive transfusion, and severe systemic illness can blur the pattern by affecting multiple hemostatic layers.

Pattern	Platelet count	PT	PTT	Fibrinogen/D-dimer	Likely diagnosis
Isolated thrombocytopenia, mucosal bleeding	Low	Normal	Normal	Usually normal	ITP, drug-induced thrombocytopenia, viral, marrow if other lines abnormal
Thrombocytopenia + thrombosis 5-10 days after heparin	Low or falling >50%	Normal	Normal or heparin effect	Normal	HIT - stop all heparin and start non-heparin anticoagulant if likely
MAHA + thrombocytopenia + normal PT/PTT	Low	Normal	Normal	Usually not DIC pattern	TTP/HUS - plasma exchange for suspected TTP
Mucosal bleeding, normal platelets, prolonged bleeding time	Normal	Normal	Normal or high	Normal	vWD or platelet dysfunction
Deep bleeding/hemarthrosis in male	Normal	Normal	High	Normal	Hemophilia A/B or factor inhibitor
Diffuse oozing in sepsis/trauma/malignancy	Low	High	High	Low fibrinogen, high D-dimer	DIC
Prolonged PT first after antibiotics/poor intake	Normal	High	May later rise	Normal	Vitamin K deficiency or warfarin effect
Prolonged PT/PTT with thrombocytopenia in cirrhosis	Low/normal	High	High	Can be low	Liver disease coagulopathy; thrombosis still possible

7. Platelet Syndromes: Low Count vs Bad Function

Thrombocytopenia is a count problem. Platelet dysfunction is a quality problem. A normal platelet count does not rule out primary hemostatic disease, especially if the patient uses aspirin/NSAIDs, has uremia, has inherited platelet adhesion/aggregation disorders, or has vWD. Conversely, a low platelet count does not automatically mean ITP; pseudo-thrombocytopenia, marrow failure, sepsis, DIC, TTP, HIT, hypersplenism, pregnancy-related disorders, and medications must remain visible.

Disorder	Core mechanism	Platelet count	Bleeding/clotting clue	Treatment logic
ITP	IgG-mediated platelet destruction	Often very low; other lines normal	Petechiae/purpura; splenomegaly absent	Steroids/IVIG for significant bleeding or very low count; thrombopoietin agonists/rituximab/splenectomy in selected chronic disease
TTP	Severe ADAMTS13 deficiency -> large vWF multimers	Low	MAHA, neuro signs, fever, renal injury; PT/PTT normal	Urgent plasma exchange + immunosuppression; do not wait for ADAMTS13 if high suspicion
HUS	Endothelial injury, often Shiga toxin or complement-mediated	Low	MAHA with renal failure prominent	Supportive care; complement blockade in atypical HUS
HIT	Anti-PF4/heparin antibodies activate platelets	Falling platelets	Thrombosis more than bleeding; onset usually days 5-10	Stop heparin, send testing, use non-heparin anticoagulant
Bernard-Soulier	GpIb defect: impaired adhesion	Mildly low; giant platelets	Mucocutaneous bleeding; abnormal ristocetin not corrected by normal plasma	Supportive; avoid antiplatelet drugs
Glanzmann	GpIIb/IIIa defect: impaired aggregation	Normal	Mucocutaneous bleeding; abnormal aggregation	Platelet transfusion or rFVIIa in severe bleeding
vWD	Reduced/defective vWF impairs adhesion and factor VIII stabilization	Usually normal	Epistaxis, menorrhagia, dental bleeding; PTT may rise	Desmopressin for responsive types; vWF/factor VIII concentrate when needed

8. Transfusion Products and Reactions

Transfusion is supportive therapy, not a diagnosis. The product must match the missing component: packed red cells for oxygen-carrying capacity, platelets for platelet number/function when bleeding or before high-risk procedures, plasma for multiple coagulation factor replacement, cryoprecipitate for fibrinogen and factor VIII/vWF-rich replacement in selected contexts, and specific concentrates or reversal agents when available. A restrictive red cell strategy is often appropriate in stable hospitalized adults, while active bleeding, shock, acute coronary syndrome, severe hypoxemia, and procedure-specific thresholds require clinical judgment.

Product	What it supplies	Typical use	Important caution
Packed RBCs	Red cell mass	Symptomatic anemia, active bleeding, oxygen delivery deficit	One unit often raises Hb about 1 g/dL in adults; do not transfuse based on number alone when stable
Platelets	Platelet number/function	Severe thrombocytopenia with bleeding; procedure prophylaxis; chemotherapy-related thrombocytopenia	Avoid routine platelet transfusion in TTP unless life-threatening bleeding because thrombosis may worsen
FFP/plasma	Broad coagulation factors	Bleeding with multiple factor deficiency, urgent warfarin reversal if PCC unavailable, massive transfusion	Requires volume; not for isolated mild INR correction without bleeding
Cryoprecipitate	Fibrinogen, factor VIII, vWF, factor XIII	Low fibrinogen with bleeding/DIC/massive transfusion	Use fibrinogen concentrate where available per institutional practice
PCC	Vitamin K-dependent factors	Urgent warfarin reversal with bleeding or emergent surgery	Thrombosis risk; give vitamin K with warfarin reversal

Reaction	Timing	Mechanism	Key findings	Action
Acute hemolytic	During or soon after	ABO incompatibility	Fever, flank pain, hypotension, hemoglobinuria, DIC	Stop transfusion, supportive care, clerical check, blood bank workup
Febrile nonhemolytic	During/soon after	Cytokines or leukocyte antibodies	Fever/chills without hemolysis	Stop and evaluate; antipyretics after serious causes excluded
Allergic/anaphylactic	Immediate	Recipient allergy; IgA deficiency in severe cases	Urticaria to hypotension/airway compromise	Antihistamines; epinephrine for anaphylaxis
TRALI	Within 6 hours	Donor antibodies activate pulmonary neutrophils	Noncardiogenic pulmonary edema, hypoxemia	Supportive respiratory care; notify blood bank
TACO	Within hours	Volume overload	Pulmonary edema, hypertension, JVD, high BNP	Slow/stop transfusion, diuretics, oxygen
Delayed hemolytic	Days-weeks	Anamnestic alloantibody	Falling Hb, jaundice, positive DAT	Supportive, antigen-negative blood in future

9. Hematologic Malignancies: Pattern Recognition

Hematologic malignancies are disorders of clonal cell expansion, impaired normal hematopoiesis, immune dysfunction, and tissue infiltration. They can present as cytopenias, leukocytosis, blasts, lymphadenopathy, splenomegaly, B symptoms, bone pain, hypercalcemia, renal failure, hyperviscosity, recurrent infections, or oncologic emergencies. The classification is not only histologic; it is also temporal. Acute leukemias progress rapidly and involve immature blasts. Chronic leukemias are often more indolent and involve more mature cells. Lymphomas present as tissue-based lymphoid tumors. Plasma cell disorders present as monoclonal immunoglobulin disease with marrow, bone, kidney, calcium, and immune consequences.

Disease	Cell/process	Classic clues	Diagnostic anchor	Emergency to remember
AML	Myeloid blasts	Older adult; anemia/infection/bleeding; Auer rods; DIC in APL	Bone marrow/flow/cytogenetics	APL requires urgent ATRA; leukostasis/tumor lysis
ALL	Lymphoblasts	Child or adult; bone pain, lymphadenopathy; mediastinal mass in T-ALL	Bone marrow/flow/cytogenetics	CNS involvement, tumor lysis, leukostasis
CML	BCR-ABL myeloproliferation	Very high WBC, left shift, basophilia, splenomegaly	Philadelphia chromosome/BCR-ABL	Blast crisis

Disease	Cell/process	Classic clues	Diagnostic anchor	Emergency to remember
CLL/SLL	Mature clonal B cells	Older adult; lymphocytosis, smudge cells, hypogammaglobulinemia	Flow cytometry	Autoimmune hemolysis, Richter transformation
Hodgkin lymphoma	Reed-Sternberg cell tumor microenvironment	Painless nodes, B symptoms, contiguous spread, mediastinal mass	Excisional node biopsy	Airway/SVC compromise if bulky mediastinal disease
Non-Hodgkin lymphoma	Diverse B/T/NK lymphoid tumors	Painless nodes or extranodal disease; variable pace	Tissue biopsy with immunophenotype	Tumor lysis, spinal/airway compression, CNS disease
Multiple myeloma	Clonal plasma cells	CRAB: hyperCalcemia, Renal injury, Anemia, Bone lesions	SPEP/UPEP, free light chains, marrow, imaging	Renal failure, hyperviscosity, spinal cord compression
Waldenstrom	Lymphoplasmacytic IgM clone	Hyperviscosity, neuropathy, bleeding, lymphadenopathy	IgM monoclonal protein + marrow	Hyperviscosity needs plasmapheresis

10. Plasma Cell Disorders and Paraprotein Logic

A monoclonal protein is a clue to clonal B-cell or plasma-cell biology, not always cancer requiring immediate therapy. MGUS has an M-protein without myeloma-defining events and is monitored. Smoldering myeloma has higher clonal burden without end-organ damage. Multiple myeloma becomes clinically significant when clonal plasma cells cause organ injury or meet biomarker-defined high-risk criteria. Renal failure may occur from light-chain cast nephropathy; bone disease reflects osteoclast activation and osteoblast suppression; anemia reflects marrow replacement and inflammation; infection reflects impaired normal immunoglobulin production.

Paraprotein evaluation

Suspect paraproteinemia with unexplained anemia, renal failure, hypercalcemia, bone pain/fractures, high total protein, low anion gap, recurrent infection, neuropathy, hyperviscosity, or rouleaux. | Order SPEP with immunofixation, serum free light chains, UPEP/urine immunofixation if needed, CBC, calcium, creatinine, albumin, LDH, beta-2 microglobulin, and imaging for lytic lesions. | Determine whether this is MGUS, smoldering myeloma, symptomatic myeloma, AL amyloidosis, Waldenstrom macroglobulinemia, or another lymphoplasmacytic process. | Treat immediately if end-organ injury is present: renal failure from light chains, hypercalcemia, spinal cord compression, pathologic fracture, severe anemia, recurrent serious infection, or hyperviscosity.

11. Solid Oncology Overview for Internal Medicine

Solid oncology in internal medicine is less about memorizing every tumor regimen and more about recognizing prevention, screening, staging, complications, treatment toxicity, and survivorship needs. Stage is determined by tumor extent, nodal involvement, and metastasis. Treatment is local, systemic, or combined: surgery and radiation primarily address local/regional disease; chemotherapy, endocrine therapy, targeted therapy, immunotherapy, and cellular therapy address systemic risk. Internal medicine clinicians often detect cancer, manage comorbidities around therapy, respond to emergencies, and distinguish treatment toxicity from progression or infection.

Clinical task	Key principle	Examples
Screening	Screen average-risk patients by guideline age/risk; modify for high-risk syndromes and prior findings	Breast, cervical, colorectal, lung; genetic risk for BRCA/Lynch; colonoscopy intervals after polyps
Staging	Stage before definitive therapy when possible; histology and biomarkers increasingly determine treatment	TNM staging, PET/CT, MRI, sentinel node biopsy, ER/PR/HER2, EGFR/ALK, MSI/MMR, PD-L1
Treatment intent	Curative, adjuvant, neoadjuvant, definitive, palliative, or supportive intent changes risk tolerance	Adjuvant chemo after surgery; neoadjuvant therapy to shrink tumor; palliative radiation for bone pain
Toxicity triage	Assume infection, thrombosis, immune toxicity, metabolic emergency, or organ toxicity until excluded	Neutropenic fever, checkpoint colitis, pneumonitis, myocarditis, TLS, hypercalcemia
Survivorship	Long-term complications persist after remission	Anthracycline cardiomyopathy, secondary malignancy, neuropathy, infertility, bone health, psychosocial burden

12. Oncologic Emergencies

Oncologic emergencies are high-yield because the first treatment may be lifesaving and often precedes final cancer histology. The emergency is defined by physiology: compression, metabolic toxicity, vascular obstruction, infection, bleeding, clotting, or treatment immune toxicity. Always ask whether the patient has new neurologic deficits, bowel/bladder dysfunction, severe back pain, facial/upper-extremity swelling, dyspnea, fever during neutropenia, severe calcium elevation, renal failure with uric acid/phosphate/potassium elevation, or rapidly rising leukocyte count with neurologic/respiratory symptoms.

Emergency	Typical clues	Mechanism	Immediate management concept
Neutropenic fever	Fever with ANC <500/uL or expected decline	Loss of neutrophil defense; infection may lack localizing signs	Prompt broad antipseudomonal antibiotics after cultures; do not delay for imaging
Tumor lysis syndrome	High K, phosphate, uric acid; low Ca; AKI	Rapid cell breakdown after therapy or high-burden tumor	IV fluids, rasburicase/allopurinol by risk, electrolyte management, dialysis if severe
Hypercalcemia of malignancy	Confusion, constipation, polyuria, dehydration, QT shortening	PTHrP, bone metastases, vitamin D-mediated lymphoma	IV fluids, calcitonin for rapid effect, IV bisphosphonate/denosumab, treat tumor
Spinal cord compression	Back pain, weakness, sensory level, bowel/bladder dysfunction	Epidural tumor compression	Steroids if suspected, urgent MRI, radiation/surgery consultation
SVC syndrome	Facial/arm swelling, venous distension, dyspnea, headache	Mediastinal mass obstructs venous return	Airway assessment, imaging, steroids/radiation/stent depending cause and severity
Leukostasis	Very high blasts with dyspnea, hypoxemia, headache, confusion	Microvascular sludging by blasts	Hydration, cytoreduction, leukapheresis in selected cases, treat leukemia
Checkpoint inhibitor toxicity	Colitis, hepatitis, dermatitis, pneumonitis, endocrinopathies, myocarditis	Immune activation against normal tissue	Hold immunotherapy, grade severity, corticosteroids for significant immune-related adverse events

13. Medication and Treatment Toxicity Integration

Cancer treatment toxicity is often pattern-based. Cytotoxic chemotherapy injures rapidly dividing tissue, leading to myelosuppression, mucositis, alopecia, nausea, diarrhea, infertility, and infection risk. Targeted therapies create pathway-specific toxicities. Immunotherapy can inflame almost any organ. Radiation toxicity follows the field of exposure and can be acute inflammatory or delayed fibrotic. The safest mental model is to connect drug class to organ risk, but also to keep infection, thrombosis, progression, and metabolic emergencies on the differential.

Therapy/class	Signature toxicity pattern	Clinical trap
Anthracyclines	Dose-related cardiomyopathy	Dyspnea after therapy may be heart failure, not only anemia or lung disease
Bleomycin	Pulmonary toxicity	Avoid dismissing cough/dyspnea; oxygen exposure may worsen risk in some contexts
Cisplatin	Nephrotoxicity, ototoxicity, neuropathy	Creatinine rise may coexist with dehydration, TLS, or myeloma kidney
Cyclophosphamide/ifosfamide	Hemorrhagic cystitis, myelosuppression	Hematuria needs medication history and malignancy/infection evaluation
Methotrexate	Mucositis, marrow suppression, hepatotoxicity, pneumonitis	Renal dysfunction increases toxicity
Taxanes/vinca alkaloids	Neuropathy; vinca can cause ileus/SIADH	Constipation/ileus may be drug effect, obstruction, or hypercalcemia
Immune checkpoint inhibitors	Colitis, hepatitis, dermatitis, pneumonitis, endocrinopathies, myocarditis	Symptoms may mimic infection or progression; high-grade toxicity often needs steroids
Radiation	Field-dependent inflammation/fibrosis; marrow suppression if large marrow field	Late toxicity can appear months-years after treatment

14. Integrated Case Bank

Case 1 - Fatigue and microcytosis

A 67-year-old man has fatigue, Hb 8.9 g/dL, MCV 72 fL, high RDW, low ferritin, and positive fecal occult blood.

Interpretation: Iron deficiency anemia from chronic GI blood loss until proven otherwise. Treat iron deficiency, but the diagnostic priority is evaluation for GI bleeding and colorectal malignancy. Do not stop at oral iron response.

Case 2 - Normal MCV but low reticulocytes

A patient with rheumatoid arthritis has Hb 10.2 g/dL, MCV 88 fL, low serum iron, low TIBC, high ferritin, and low reticulocyte index.

Interpretation: Anemia of inflammation. Hepcidin traps iron in macrophages and suppresses erythropoiesis. Treat underlying inflammation; iron is not automatically helpful unless true iron deficiency coexists.

Case 3 - Macrocytosis and neurologic symptoms

An older patient has paresthesias, gait instability, glossitis, MCV 115 fL, hypersegmented neutrophils, high MMA, and high homocysteine.

Interpretation: Vitamin B12 deficiency. Treat promptly with B12; folate alone may correct anemia while neurologic injury progresses.

Case 4 - Hemolysis after antibiotics

A patient develops jaundice, dark urine, back pain, low haptoglobin, high LDH, and bite cells after an oxidant medication.

Interpretation: G6PD-associated oxidative hemolysis. Stop trigger, provide supportive care, and remember testing during the acute event may be falsely normal if older deficient RBCs have already hemolyzed.

Case 5 - Thrombocytopenia with normal PT/PTT

A patient has confusion, fever, anemia, platelets 18,000/uL, schistocytes, elevated LDH, indirect bilirubin, creatinine 1.7 mg/dL, normal PT/PTT.

Interpretation: TTP until proven otherwise. Start plasma exchange and immunosuppression urgently; do not wait for ADAMTS13 testing if suspicion is high.

Case 6 - Platelets fall after heparin

A hospitalized patient on heparin develops a >50% platelet fall on day 7 and new DVT.

Interpretation: HIT. Stop heparin, avoid platelet transfusion unless severe bleeding, test for PF4/heparin antibodies with confirmatory functional assay when needed, and anticoagulate with a non-heparin agent.

Case 7 - Hemarthrosis in a boy

A male child has recurrent knee hemarthrosis, normal platelet count, normal PT, prolonged PTT, and low factor VIII.

Interpretation: Hemophilia A. Deep joint bleeding points to coagulation factor deficiency, not platelet disease. Treat with factor replacement; desmopressin may help mild disease.

Case 8 - Sepsis with oozing

A septic patient has diffuse line oozing, platelets 40,000/uL, prolonged PT/PTT, high D-dimer, low fibrinogen, and schistocytes.

Interpretation: DIC. Treat the underlying sepsis and provide supportive blood products if bleeding or procedures require correction.

Case 9 - Bone pain and renal failure

A 72-year-old has back pain, anemia, creatinine elevation, calcium 12.3 mg/dL, rouleaux, and an M-spike.

Interpretation: Multiple myeloma. Evaluate with SPEP/UPEP, immunofixation, free light chains, marrow, and skeletal imaging. Watch for renal failure, fracture, infection, hyperviscosity, and cord compression.

Case 10 - Bulky mediastinal mass

A young adult has painless cervical nodes, pruritus, night sweats, and a mediastinal mass.

Interpretation: Hodgkin lymphoma is likely. Excisional lymph node biopsy is preferred; staging determines therapy. Evaluate airway/SVC risk in bulky mediastinal disease.

Case 11 - Febrile neutropenia

A chemotherapy patient has ANC 200/uL and fever 38.6 C without localizing symptoms.

Interpretation: Neutropenic fever. Obtain cultures, but give empiric antipseudomonal antibiotics promptly. A lack of focal findings does not reassure because neutrophils are needed to localize infection.

Case 12 - Hyperkalemia after chemotherapy

A patient with bulky lymphoma starts treatment and develops potassium 6.2, phosphate 7.0, uric acid elevation, low calcium, and AKI.

Interpretation: Tumor lysis syndrome. Aggressive IV fluids, urate-lowering therapy by risk/severity, electrolyte management, telemetry, and nephrology involvement if severe.

15. High-Yield Comparison Tables

A capstone chapter should make contrasts explicit. The following tables are designed for fast retrieval and graph extraction: they encode entities, mechanisms, diagnostic anchors, and management links in a compact format.

Syndrome pair	Shared feature	Best separator
TTP vs DIC	Thrombocytopenia, anemia, schistocytes	TTP usually has normal PT/PTT/fibrinogen; DIC has prolonged PT/PTT, low fibrinogen, high D-dimer
ITP vs TTP	Low platelets	ITP is isolated thrombocytopenia; TTP has MAHA, neurologic/renal features, fever, and schistocytes

Syndrome pair	Shared feature	Best separator
vWD vs hemophilia	Bleeding disorder with possible prolonged PTT	vWD causes mucosal bleeding and abnormal platelet adhesion; hemophilia causes deep bleeding/hemarthrosis
Iron deficiency vs thalassemia	Microcytosis	Iron deficiency: low ferritin/high RDW; thalassemia: normal/high iron, target cells, family background, microcytosis disproportionate
B12 vs folate deficiency	Megaloblastic macrocytosis	B12: neurologic symptoms and high MMA; folate: no neurologic injury and normal MMA
Leukemoid reaction vs CML	High WBC with myeloid cells	Leukemoid: infection/inflammation and high LAP; CML: BCR-ABL, basophilia, splenomegaly
MGUS vs multiple myeloma	Monoclonal protein	Myeloma has end-organ damage or myeloma-defining biomarkers; MGUS lacks CRAB/myeloma-defining events
Hodgkin vs NHL	Lymphadenopathy/B symptoms	Hodgkin often contiguous spread with Reed-Sternberg cells; NHL is more diverse, often extranodal or noncontiguous

16. Diagnostic Algorithms

Thrombocytopenia algorithm

Confirm true low platelets: repeat CBC, smear for platelet clumping, review prior counts. | Check whether other cell lines are abnormal. Pancytopenia suggests marrow failure/infiltration, hypersplenism, severe infection, or nutritional/toxic causes. | If isolated: review medications, alcohol, viral illness, pregnancy, liver/spleen disease; consider ITP if no secondary cause. | If thrombosis or heparin exposure: calculate clinical HIT probability and stop heparin if intermediate/high suspicion. | If anemia + schistocytes: evaluate MAHA. Normal PT/PTT suggests TTP/HUS; abnormal coagulation with low fibrinogen suggests DIC. | Treat based on danger: bleeding, platelet count <10,000/uL, CNS/GI bleeding, TTP, HIT, DIC, or urgent procedure.

Lymphadenopathy algorithm

Identify localized vs generalized nodes, duration, size, tenderness, mobility, supraclavicular location, B symptoms, hepatosplenomegaly, infectious exposure, autoimmune clues, and medication history. | Short-lived tender nodes with infection signs can be observed or treated as infection; persistent, firm, supraclavicular, generalized, or B-symptom nodes require tissue diagnosis. | Prefer excisional biopsy when lymphoma is suspected because architecture matters. Fine-needle aspiration may miss lymphoma subtype and staging information. | Stage with history, exam, labs, imaging, and marrow evaluation when indicated. Urgent complications include airway compromise, SVC syndrome, spinal cord compression, tumor lysis, and cytopenias.

Cancer patient with fever

Check ANC and recent chemotherapy/immunotherapy. ANC <500/uL or expected decline with fever is neutropenic fever. | Culture blood/urine and image based on symptoms, but give empiric antipseudomonal antibiotics promptly for neutropenic fever. | Consider noninfectious causes: tumor fever, transfusion reaction, drug fever, thrombosis/PE, adrenal insufficiency, checkpoint inhibitor inflammation. | Search for high-risk infection sites: catheter, lung, GI mucositis/typhlitis, skin/soft tissue, urinary tract, perianal region, CNS when symptomatic.

17. Graph-RAG Extraction Map

The following extraction map is designed for a graph database. Each row can be converted into concept nodes and relationship edges. The most useful edges in heme/onc are: causes, presents_with, diagnosed_by, treated_by, complicated_by, differentiates_from, contraindicates, requires_urgent_treatment, and associated_with.

Node	Class	Core properties	Suggested edges
Anemia	Syndrome	Low Hb/Hct; symptoms depend on acuity and cardiopulmonary reserve	classified_by -> MCV; classified_by -> reticulocyte index; treated_by -> transfusion when unstable/severe
Iron deficiency anemia	Microcytic anemia	Low ferritin, low iron, high TIBC, high RDW	caused_by -> chronic blood loss; diagnosed_by -> iron studies; requires_workup_for -> GI malignancy in men/postmenopausal women
B12 deficiency	Macrocytic anemia	Megaloblastic anemia, neurologic injury, high MMA	treated_by -> cobalamin; differentiates_from -> folate deficiency; associated_with -> pernicious anemia

Node	Class	Core properties	Suggested edges
Hemolysis	RBC destruction syndrome	High LDH, high indirect bilirubin, low haptoglobin, reticulocytosis	classified_by -> smear; diagnosed_by -> DAT when immune suspected; complicated_by -> pigment gallstones
Sickle cell disease	Hemoglobinopathy	HbS polymerization, vaso-occlusion, chronic hemolysis	complicated_by -> acute chest syndrome; treated_by -> hydroxyurea; treated_by -> transfusion/exchange in severe complications
TTP	Thrombotic microangiopathy	ADAMTS13 deficiency, MAHA, thrombocytopenia, normal PT/PTT	requires_urgent_treatment -> plasma exchange; differentiates_from -> DIC; presents_with -> neurologic symptoms
HIT	Immune thrombotic thrombocytopenia	Anti-PF4/heparin antibodies; thrombosis; platelet fall	caused_by -> heparin; treated_by -> non-heparin anticoagulant; contraindicates -> heparin
DIC	Consumptive coagulopathy	Low platelets, high PT/PTT, high D-dimer, low fibrinogen	caused_by -> sepsis/trauma/malignancy/obstetric complication; treated_by -> underlying cause; treated_by -> blood products if bleeding
Multiple myeloma	Plasma cell malignancy	CRAB, M protein, clonal plasma cells, lytic lesions	diagnosed_by -> SPEP/UPEP/free light chains/marrow; complicated_by -> renal failure; complicated_by -> hypercalcemia
Hodgkin lymphoma	Lymphoma	Reed-Sternberg cells, contiguous spread, B symptoms	diagnosed_by -> excisional node biopsy; staged_by -> Ann Arbor; treated_by -> chemo/radiation depending stage
AML	Acute leukemia	Myeloid blasts; Auer rods; marrow failure	complicated_by -> leukostasis; complicated_by -> tumor lysis; subtype_APL treated_by -> ATRA
Neutropenic fever	Oncologic emergency	Fever with severe neutropenia	requires_urgent_treatment -> antipseudomonal antibiotics; caused_by -> chemotherapy/marrow failure
Tumor lysis syndrome	Oncologic emergency	HyperK, hyperphos, hyperuricemia, hypocalcemia, AKI	caused_by -> rapid tumor cell breakdown; treated_by -> IV fluids; treated_by -> rasburicase/allopurinol

18. Practice Questions

Question 1

A woman with fatigue has Hb 9.5, MCV 68, ferritin 5, high TIBC, and high RDW. What is the mechanism and what must be excluded if she is postmenopausal?

Interpretation: Iron deficiency from insufficient iron for heme synthesis. In a postmenopausal patient, occult GI blood loss and colorectal malignancy must be excluded.

Question 2

A man has jaundice, anemia, high LDH, low haptoglobin, spherocytes, and a positive DAT. What is the diagnosis category?

Interpretation: Warm autoimmune hemolytic anemia. Spherocytes plus positive DAT indicate immune-mediated RBC destruction.

Question 3

A hospitalized patient has schistocytes, thrombocytopenia, prolonged PT/PTT, low fibrinogen, high D-dimer, and bleeding from IV sites. What syndrome is most likely?

Interpretation: DIC, most likely triggered by severe systemic illness such as sepsis, trauma, malignancy, or obstetric catastrophe.

Question 4

Why is platelet transfusion usually avoided in TTP?

Interpretation: TTP is a microthrombotic process; adding platelets may fuel thrombosis unless there is life-threatening bleeding.

Question 5

A patient has epistaxis, menorrhagia, normal platelet count, normal PT, and prolonged PTT. Which disorder links platelet adhesion and factor VIII stability?

Interpretation: von Willebrand disease.

Question 6

Which plasma cell disorder is associated with IgM, hyperviscosity, lymphadenopathy, splenomegaly, and neuropathy?

Interpretation: Waldenstrom macroglobulinemia.

Question 7

A cancer patient with ANC 100/uL and fever should receive what time-sensitive treatment?

Interpretation: Prompt empiric antipseudomonal antibiotics after appropriate cultures; treatment should not be delayed for complete localization.

Question 8

A patient with back pain, leg weakness, and urinary retention in known metastatic cancer needs what immediate diagnostic and management steps?

Interpretation: Suspect malignant spinal cord compression; give steroids if clinically suspected and obtain urgent MRI with radiation/neurosurgical consultation.

19. Common Pitfalls

- Assuming normal MCV rules out nutritional anemia. Mixed deficiencies or early disease may normalize MCV while RDW rises.
- Treating iron deficiency in an older adult without evaluating for occult blood loss.
- Calling thrombocytopenia ITP before reviewing smear, medication exposures, heparin timing, hemolysis labs, liver/spleen disease, and other cell lines.
- Waiting for ADAMTS13 before treating high-suspicion TTP.
- Confusing TTP with DIC: DIC consumes coagulation factors; TTP usually has normal PT/PTT.
- Transfusing platelets reflexively in TTP or HIT without severe bleeding.
- Assuming elevated INR in cirrhosis means the patient is auto-anticoagulated. Cirrhosis can have bleeding and thrombosis risk.
- Using folate alone in a patient who may have B12 deficiency.
- Missing myeloma in anemia plus renal failure, hypercalcemia, bone pain, or high total protein.
- Delaying antibiotics in neutropenic fever because the patient looks well or has no localizing signs.
- Treating cancer-related back pain as routine pain when neurologic deficits or bladder symptoms suggest cord compression.
- Attributing dyspnea after cancer therapy only to anemia; consider PE, infection, heart failure, pneumonitis, drug toxicity, effusion, and progression.

20. Selected Clinical Cross-Check References

These references were used for clinical anchoring and safety framing. The chapter remains an original Medvale study resource and does not reproduce source text or copyrighted images.

- AABB International Guidelines, 2023: restrictive red blood cell transfusion strategies, often using a 7 g/dL threshold in stable hospitalized adults with clinical context.
- American Society of Hematology sickle cell disease guideline resources: transfusion support, pain, cardiopulmonary/kidney complications, and health maintenance guidance.
- American Society of Hematology HIT guideline resources: clinical probability assessment, heparin cessation, and non-heparin anticoagulation principles.
- ISTH guidance on thrombotic thrombocytopenic purpura: urgent plasma exchange-based management for high-suspicion immune TTP.
- NCI PDQ and NCCN guideline resources: leukemia, lymphoma, multiple myeloma, solid tumor screening/staging/treatment, supportive care, and oncologic emergencies.
- USPSTF cancer screening recommendations: breast, cervical, colorectal, and lung screening anchors for average-risk patients, modified by individual risk.